



5365 - Identifying, counting, and mapping YSOs in Sgr B2: our Galaxy's most massive molecular cloud

Cycle: 3, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Sgr B2 NIRCAM	NIRCam Imaging	(1) SGRB2_NIRCAM
	2	Sgr B2 MIRI	MIRI Imaging	(2) SGRB2_MIRI
	999	Sgr B2 MIRI_safing_re do	MIRI Imaging	(2) SGRB2_MIRI
	3	NIRCam Schedule Buff er And Dark	NIRCam Dark	NONE
Observation Folder				

Folder	Observation	Label	Observing Template	Science Target
	998	Sgr B2 MIRI_skipped_redo	MIRI Imaging	(2) SGRB2_MIRI

ABSTRACT

In the central few hundred parsecs of our Galaxy is the gas-rich, extremely dense Central Molecular Zone (CMZ), which exhibits physical conditions unique with in the Local Group that are analogous to galactic environments at Cosmic Noon. Star formation in the CMZ is suppressed by more than an order of magnitude compared to predictions based on Galactic disk star formation laws. Within the CMZ, there is one region that stands out above the rest and accounts for more than half of the total star formation, Sgr B2. We propose to observe the forming young stellar population in Sgr B2 using a range of filters on NIRCam and MIRI that sample YSO tracers, such as accretion, presence of ices, or dust excess. We will map out where YSOs are forming to assess whether star formation occurs above different thresholds in the CMZ, to measure the overall star formation rate, and to determine whether the stellar initial mass function (IMF) is consistent with standard models. There is some evidence from ALMA observations that both the thresholds and the IMF are different in the CMZ, however measurements of the low-mass YSO population, only available with JWST, are needed to test this. The proposed observations enable a robust test of how star formation varies within the extreme environment of our Galactic Center, and will serve as a crucial template for interpreting extragalactic observations.

OBSERVING DESCRIPTION

We aim to map out the Sgr B2 molecular cloud with NIRCam and MIRI.

For NIRCam, we select a set of eleven filters: four narrow-band (PaA 1.87 micron, BrA 4.05 micron, H2 2.12 micron, and CO 4.66 micron), six medium-band (F210M, F182M, F300M, F360M, F410M, and F480M), and one wide-band F150W. MEDIUM2 readout mode is selected to be sensitive to YSOs with luminosities larger than 1 solar luminosity and to decrease the data rate. For the narrow-band PaA and BrA filters, MEDIUM8 is used to improve sensitivity and limit data rate. The narrow- and medium-band filters are selected to mitigate saturation, since the starfield in the inner galaxy is so rich that a substantial fraction (1-5%) of the target area saturates at frame 0 (<1%) or in the final frame (up to 5%). The medium-band filters will also serve as off-band continuum filters for the narrow-band filters. The narrow-band filters are used to measure specific emission features in accreting YSOs (PaA and BrA), CO and water ice in absorption to YSOs, and outflows from YSOs (H2). The wide-band filter is selected at 150 microns because the deeper measurement of the short-wavelength part of the SED deeper into the extincted cloud is valuable for measuring extinction and for spectral classification.

For MIRI, we choose three filters: F770M that includes PAHs present in YSOs, F1220W that traces the far-IR bump, and F2250W to distinguish between dust-rich YSOs and embedded giant stars. The inclusion of MIRI filters will allow for a better SED fitting to YSO models.

We adopted the standard dither patterns for imaging to optimize spatial resolution: full dither with six primary and four sub-pixel dithers for NIRCam

JWST Proposal 5365 (Created: Wednesday, May 28, 2025, 11:00:11AM Eastern Standard Time) - Overview

and full subarray with 4-point-set dithers for MIRI. The MIRI mosaic is set up to spatially match NIRCcam footprint.

The naturally allowed position angles are well-aligned with the molecular cloud, so we do not impose any special restrictions on the observations.

Proposal 5365 - Targets - Identifying, counting, and mapping YSOs in Sgr B2: our Galaxy's most massive molecular cloud

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SGRB2_NIRCAM	RA: 17 47 20.3166 (266.8346525d) Dec: -28 23 52.32 (-28.39787d) Equinox: J2000		
	Comments: Category=Stellar Cluster Description=[Young star clusters] Extended=YES				
Fixed Targets	(2)	SGRB2_MIRI	RA: 17 47 19.9748 (266.8332283d) Dec: -28 23 50.73 (-28.39742d) Equinox: J2000		
	Comments: Category=Stellar Cluster Description=[Young star clusters] Extended=YES				

Proposal 5365 - Observation 1 - Identifying, counting, and mapping YSOs in Sgr B2: our Galaxy's most massive molecular cloud

Observation	Proposal 5365, Observation 1: Sgr B2 NIRCAM									Wed May 28 16:00:11 GMT 2025	
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over lower threshold (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(1)	SGRB2_NIRCAM	RA: 17 47 20.3166 (266.8346525d) Dec: -28 23 52.32 (-28.39787d) Equinox: J2000 Comments: Category=Stellar Cluster Description=[Young star clusters] Extended=YES								
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module Gap				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		6		STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F212N	F466N+F444W	SHALLOW2	3	1	24	24	3092.19	177399	
	2	F187N	F405N+F444W	MEDIUM8	4	1	24	24	9791.934	177399	
	3	F187N	F300M	SHALLOW2	3	1	24	24	3092.19	177399	
	4	F210M	F480M	SHALLOW2	3	1	24	24	3092.19	177399	
	5	F182M	F410M	SHALLOW2	3	1	24	24	3092.19	177399	
	6	F150W	F360M	SHALLOW2	3	1	24	24	3092.19	177399	
Special Requirements	Between Dates 02-SEP-2024:00:00:00 and 09-SEP-2024:00:00:00 3 After 1 by 6 Hours to 2 Days, Exclusive Use Of Instrument										

Proposal 5365 - Observation 2 - Identifying, counting, and mapping YSOs in Sgr B2: our Galaxy's most massive molecular cloud

Observation	Proposal 5365, Observation 2: Sgr B2 MIRI										Wed May 28 16:00:11 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	SGRB2_MIRI	RA: 17 47 19.9748 (266.8332283d)								
			Dec: -28 23 50.73 (-28.39742d)								
			Equinox: J2000								
Template	Comments:										
	Category=Stellar Cluster										
	Description=[Young star clusters]										
	Extended=YES										
Mosaic	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	1	4	10.0		10.0		0.0		0.0		DEFAULT
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	309	5		5	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F770W	FASTR1	8	2	1	Dither 1	5	10	235.878	177399
	2	F1280W	FASTR1	8	2	1	Dither 1	5	10	235.878	177399
	3	F2550W	FASTR1	8	2	1	Dither 1	5	10	235.878	177399

Proposal 5365 - Observation 2 - Identifying, counting, and mapping YSOs in Sgr B2: our Galaxy's most massive molecular cloud	
Special Requirements	Group Visits within 53.0 Days Visits Same PA

Proposal 5365 - Observation 999 - Identifying, counting, and mapping YSOs in Sgr B2: our Galaxy's most massive molecular cloud

Observation	Proposal 5365, Observation 999: Sgr B2 MIRI_safing_redo										Wed May 28 16:00:11 GMT 2025
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 999:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	SGRB2_MIRI	RA: 17 47 19.9748 (266.8332283d) Dec: -28 23 50.73 (-28.39742d) Equinox: J2000								
Template	Comments: Category=Stellar Cluster Description=[Young star clusters] Extended=YES										
	Subarray										
Mosaic	FULL										
	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	1	4	10.0		10.0		0.0		0.0		ROW_ORDER
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	309	5		5	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F770W	FASTR1	8	2	1	Dither 1	5	10	235.878	177399
	2	F1280W	FASTR1	8	2	1	Dither 1	5	10	235.878	177399
	3	F2550W	FASTR1	8	2	1	Dither 1	5	10	235.878	177399

Proposal 5365 - Observation 3 - Identifying, counting, and mapping YSOs in Sgr B2: our Galaxy's most massive molecular cloud

Observation	Proposal 5365, Observation 3: NIRCam Schedule Buffer And Dark							Wed May 28 16:00:11 GMT 2025						
	Diagnostic Status: Warning													
	Observing Template: NIRCam Dark													
	Comments: Six-hour schedule buffer, terminated by dark observation, to mitigate expected long-term persistence after Obs 1.													
	No use of NIRCam during scheduling buffer is allowed.													
Diagnostics	Observations from other SIs can be scheduled during buffer.													
	(NIRCam Schedule Buffer And Dark (Obs 3)) Warning (Form): PARALLEL requirement expected. (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Template	Module		Subarray		Science Template		Occulting Mask		No. of Output Channels					
	ALL		FULL		Imaging				4					
Spectral Elements	#	Readout Pattern		Groups/Int	Integrations/Exp		Exposures/Dith		Total Integrations		Total Exposure Time		ETC Wkbk.Calc ID	
	1	MEDIUM8		10	1		1		1		1052.203			
Special Requirements	Between Dates 02-SEP-2024:00:00:00 and 09-SEP-2024:00:00:00													
	3 After 1 by 6 Hours to 2 Days, Exclusive Use Of Instrument													

Proposal 5365 - Observation 998 - Identifying, counting, and mapping YSOs in Sgr B2: our Galaxy's most massive molecular cloud

Observation	Proposal 5365, Observation 998: Sgr B2 MIRI_skipped_redo										Wed May 28 16:00:11 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 998:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	SGRB2_MIRI	RA: 17 47 19.9748 (266.8332283d)								
			Dec: -28 23 50.73 (-28.39742d)								
			Equinox: J2000								
Template	Comments:										
	Category=Stellar Cluster										
	Description=[Young star clusters]										
	Extended=YES										
Mosaic	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	1	4	10.0		10.0		0.0		0.0		ROW_ORDER
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	309	5		5	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F770W	FASTR1	8	2	1	Dither 1	5	10	235.878	177399
	2	F1280W	FASTR1	8	2	1	Dither 1	5	10	235.878	177399
	3	F2550W	FASTR1	8	2	1	Dither 1	5	10	235.878	177399

Special Requirements	Offset 2.616154422283477 arcsec, -4.921579180781649 arcsec
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